INTEROFFICE MEMORANDUM

To: CISO

From: Team 02

Date: October 24, 2026

Subject: Perimeter Firewall Configuration

Summary:

This memo outlines the efforts taken by team02 to configure the perimeter firewall to only allow expected traffic.

Palo Alto Firewall Configuration

The Palo Alto perimeter firewall is configured to only allow expected outbound and inbound packets, being implemented at the application-layer rather than layer-3 meaning that the firewall rules are configured to accept or deny packets based on the service they are attempting to access rather than the port that they are requesting.

Previous Configurations

The initial configurations for the perimeter firewall were extremely insecure, specifically allow all traffic inbound and all traffic outbound. This configuration was extremely insecure as it all all traffic to any port rather than only the expect ports and services which means malicious traffic can easily enter and penetrate our network. The screenshot below shows the initial configurations.

				Source				Destination			
	NAME	TAGS	TYPE	ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE	APPLI
1	Permit Outbound	none	interzone	any	any	any	any	any	any	any	any
2	Permit Inbound	none	interzone	external	any	any	any	any	any	any	any
3	intrazone-default	none	intrazone	any	any	any	any	(intrazone)	any	any	any
4	interzone-default	none	interzone	any	any	any	any	any	any	any	any

Current Configurations

The current configurations are allowing only the applications necessary for our business operations. For instance, rules allowing Webmail traffic, SSH, DNS, WinRM, Ping, HTTP, HTTPS, MsSQL, SMB, Splunk, IMAP, and POP3 traffic. These are bound to the specific hosts that require these.

3	AD	none	interzone	external	any	any	any	internal	AD	any	dns
4	Block everything to ...	none	universal	external	any	any	any	internal	AD	any	any
5	IPS	none	universal	any	any	any	any	internal	any	any	any
6	Rocky Ecom	none	universal	external	any	any	any	internal	Ecom	any	ping ssh web-browsing
7	Ubuntu Email	none	universal	external	any	any	any	internal	Ubuntu Email	any	imap ping pop3 roundcube smtp ssh web-browsing
8	MSSQL	none	universal	external	any	any	any	internal	MSSQL	any	ms-ds-smb ms-rdp mssql-db ping
9	Splunk	none	universal	external	any	any	any	internal	Splunk	any	ping splunk ssh web-browsing
10	Block All	none	universal	external	any	any	any	internal	any	any	any

IPS Implementation

Current firewall configurations likewise block CVEs marked medium, high, critical and capture packets through IPS. The following screenshot shows our implementation.

☑	block alot		Rules: 1	All High	any	any	medium,high,cri...	drop	extended-capture

Logging Dropped Packets

We are logging dropped packets in order to identify patterns of malicious behavior. Logging Dropped Packets was accomplished in the Palo Alto settings by going to Polices > Security, and creating a new default-deny firewall policy that drops all packets that do not match a configured accept rule and then by clicking the box for ‘Log at Session End’. This screenshot likewise shows packets being dropped.

At the time of writing no packets have been dropped yet.

		RECEIVE TIME	TYPE	FROM ZONE	TO ZONE	SOURCE	SOURCE USER	SOURCE DYNAMIC ADDRESS GROUP	DESTINATION	DESTINATION DYNAMIC ADDRESS GROUP	DYNAMIC USER GROUP	TO PORT	APPLICATION	ACTION
		10/24 06:56:15	end	external	internal	10.16.0.69			172.20.240.10			22	ssh	allow
		10/24 06:56:15	end	internal	external	172.20.240.100			1.1.1.1			53	dns-base	allow
		10/24 06:56:10	end	internal	external	172.20.240.100			1.1.1.1			53	dns-base	allow
		10/24 06:56:10	end	internal	external	172.20.240.113			23.33.29.215			443	ssl	allow
		10/24 06:56:00	end	internal	external	172.20.240.100			1.1.1.1			53	dns-base	allow

☑	block alot		Rules: 1	All High	any	any	medium,high,cri...	drop	extended-capture

If there are any concerns, questions, or clarifications, please do not hesitate to reach out.

With Regards,
Team 07